

CTRL + Incision – Medical Imaging Scan to 3D Volume Pipeline

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Introduction



2D Frame Limitations:

- Hard 3D mental reconstruction.
- Misses surrounding structure views.
- Costly, bulky, specialized software.

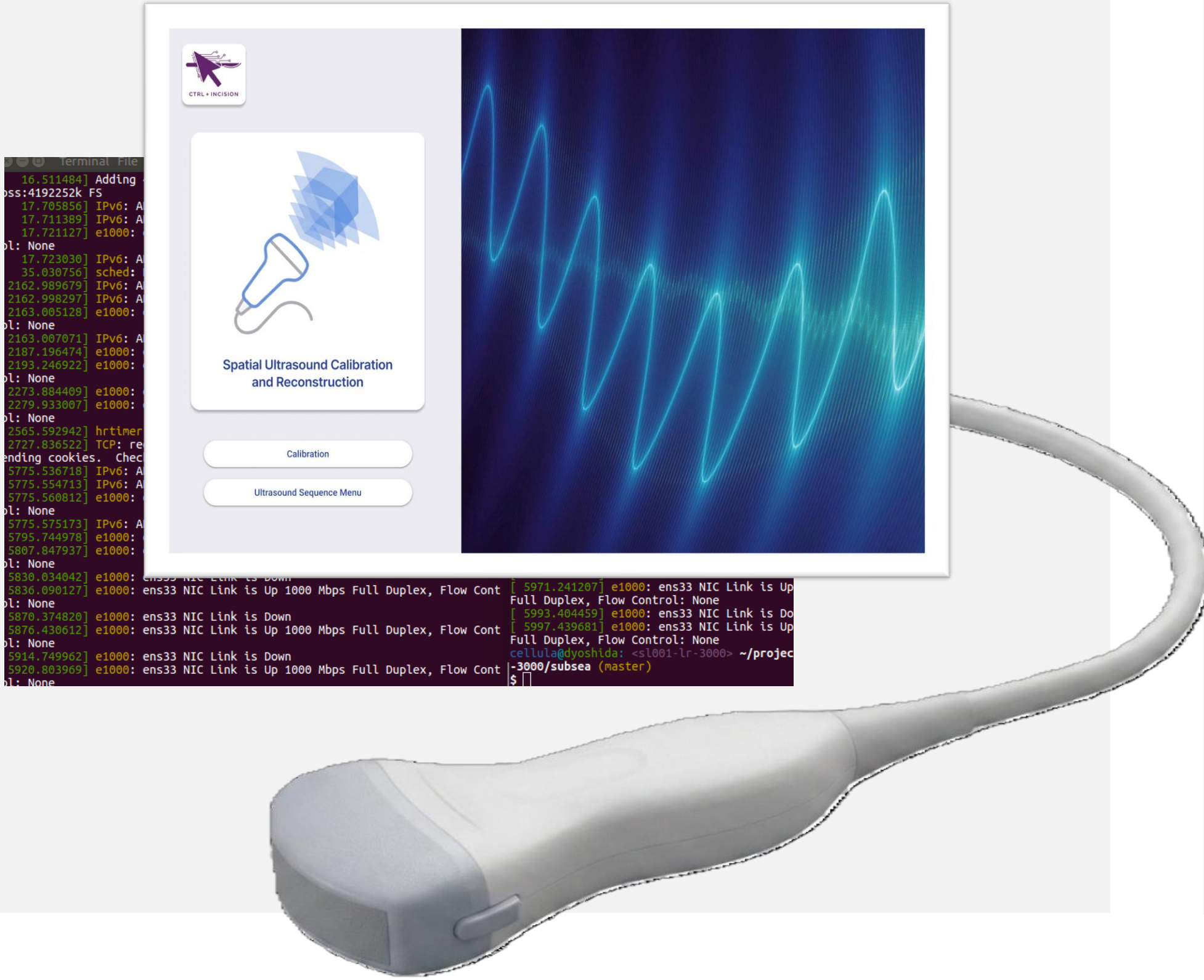


3D Reconstruction Solution:

- Precise frame coordinates.
- Coordinated surgical guidance.
- Lightweight, customizable, open-source alternative.

Motivation

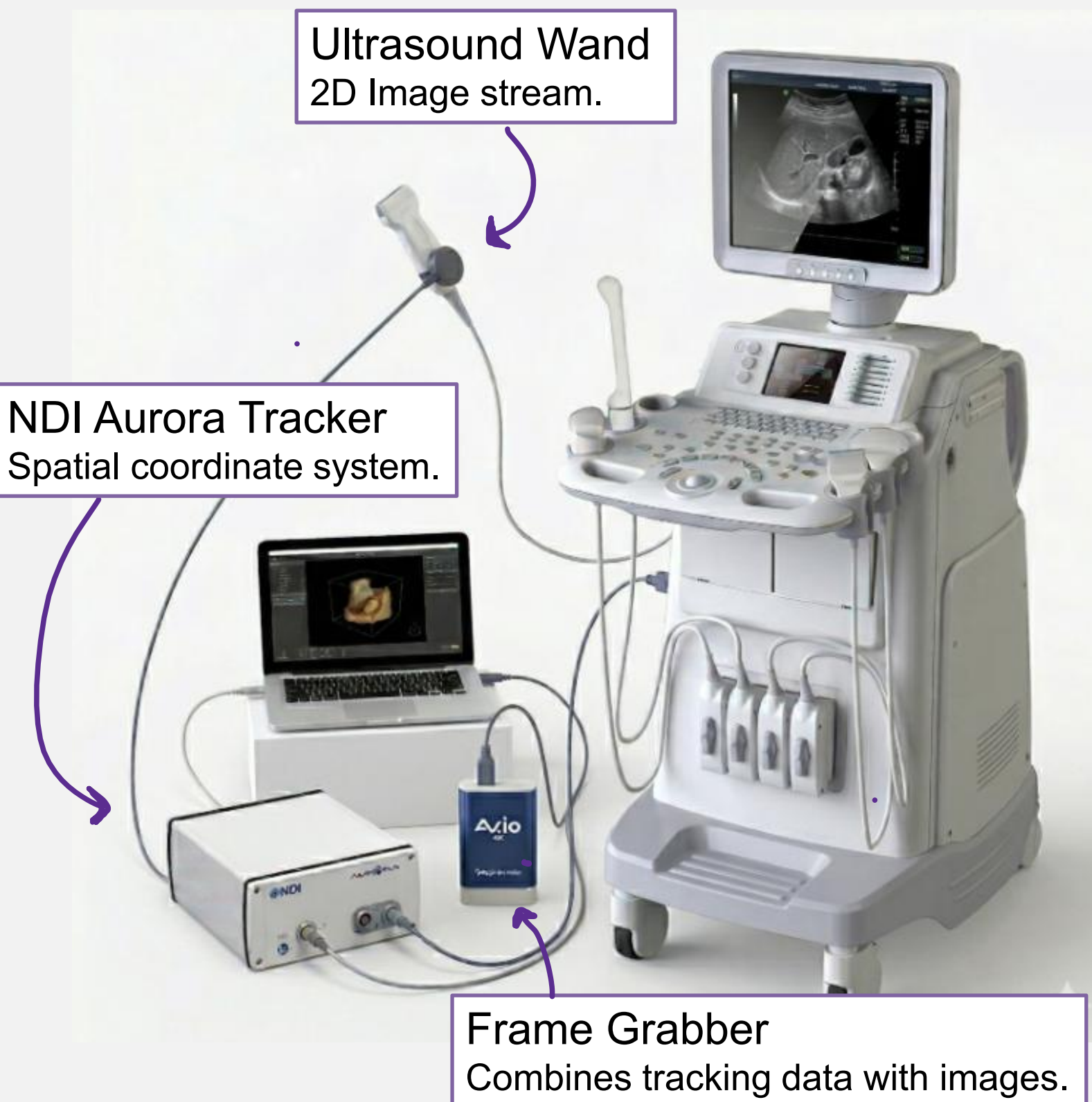
- **Software Bridge:** Simplified UI replaces complex terminal-based scripts.
- **Research Access:** Low-cost, open-source alternative to clinical suites.
- **Modular Control:** Direct hardware access without software "bloat".
- **Efficiency:** Streamlined "Calibrate -> Acquire -> Reconstruct" pipeline.



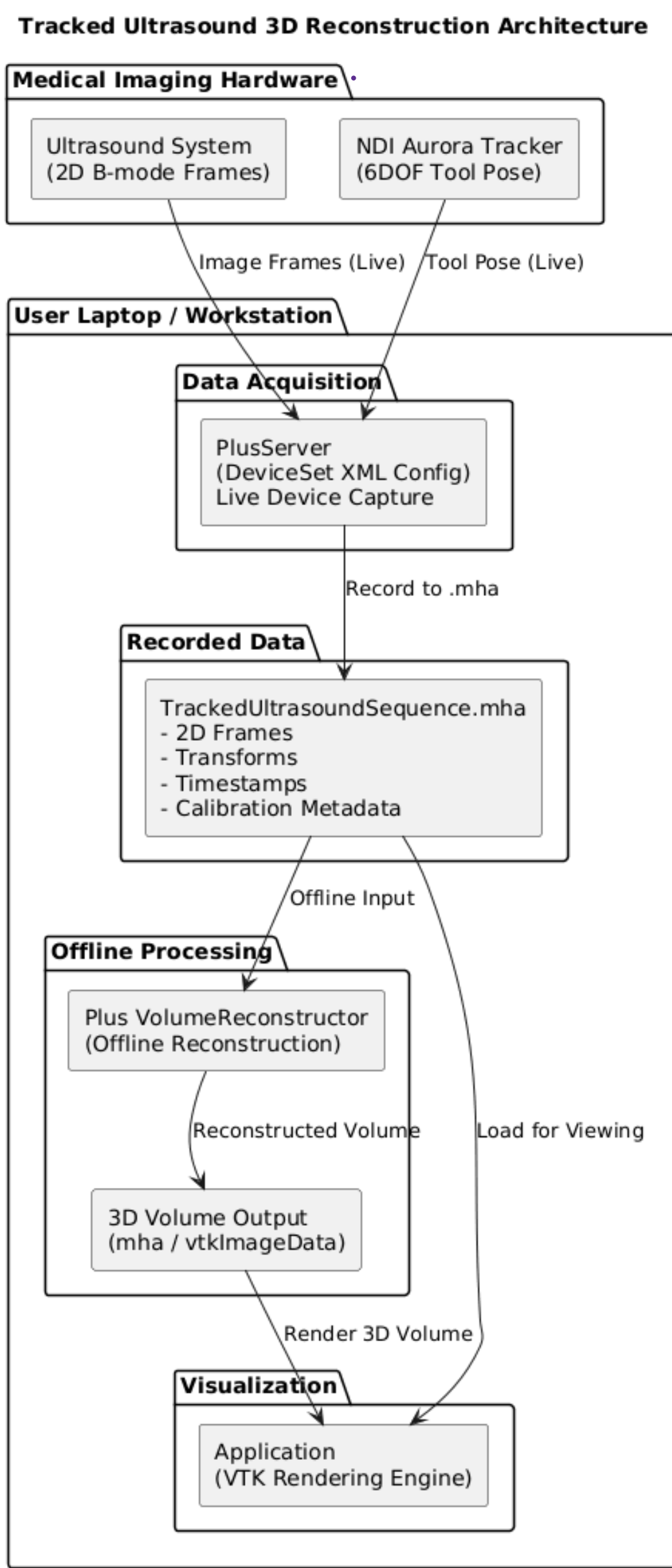
Methods

System Architecture

Hardware



Software



Methods

Code Flow Diagram

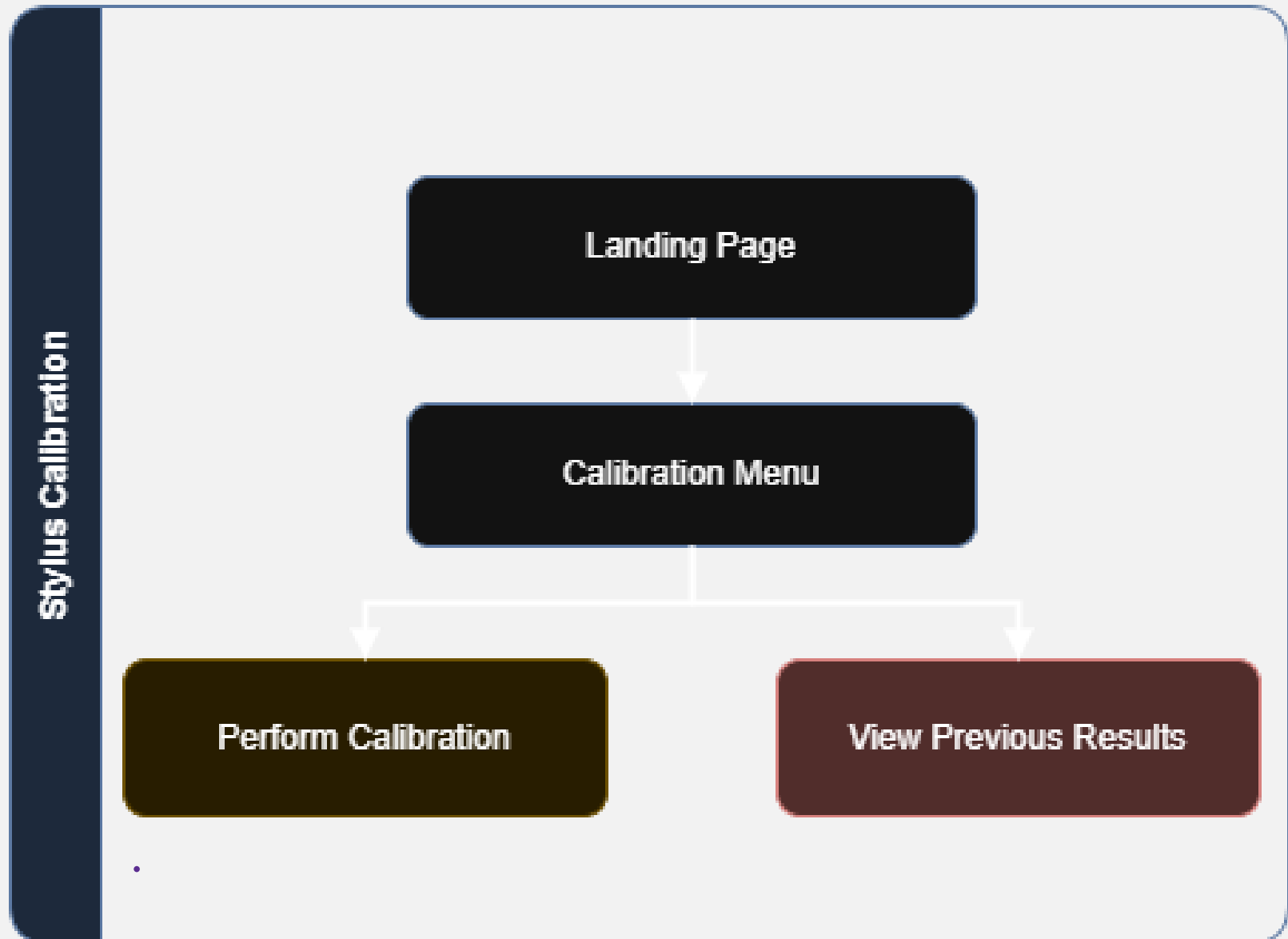


Figure 1: Calibration Code Flow

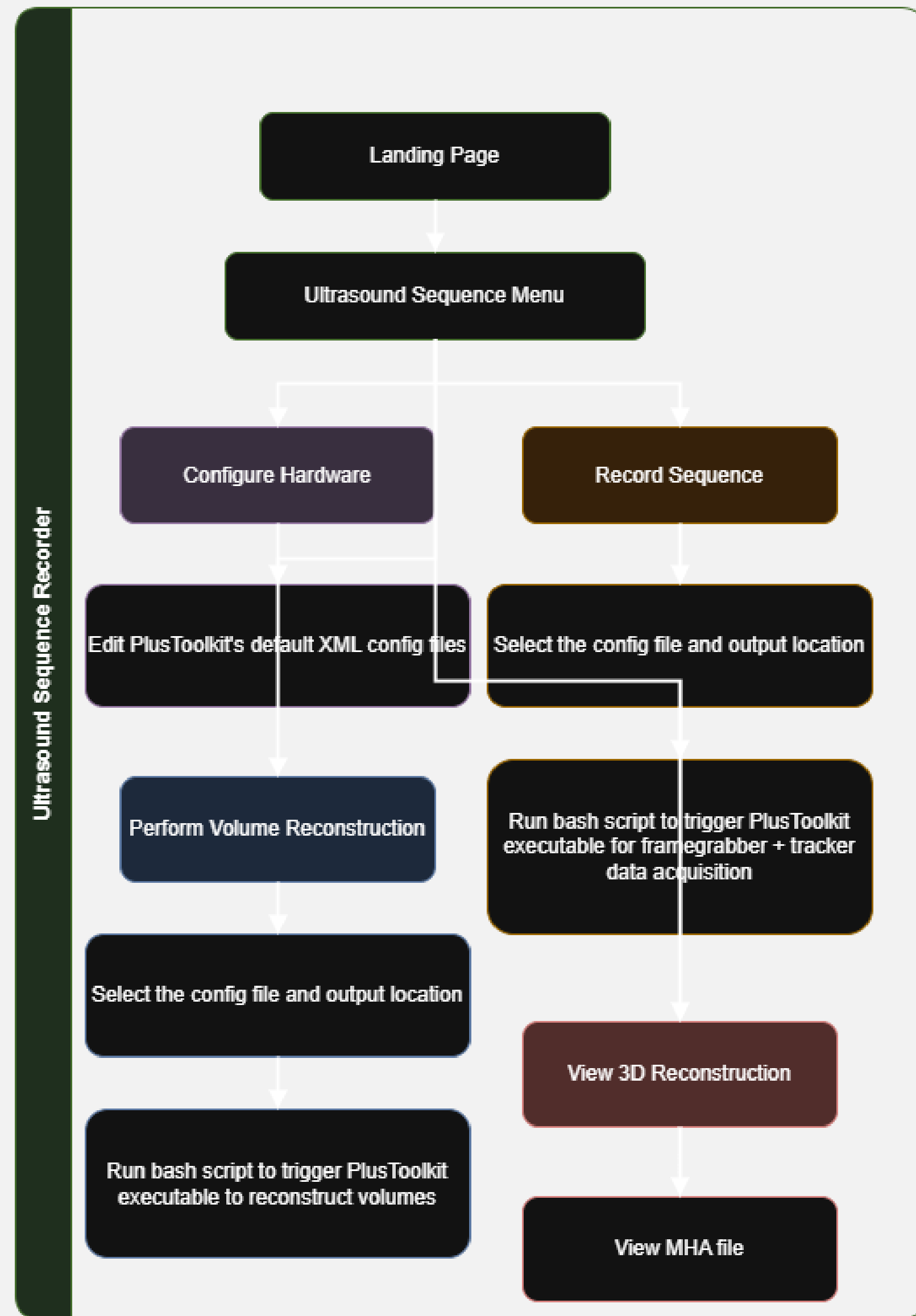
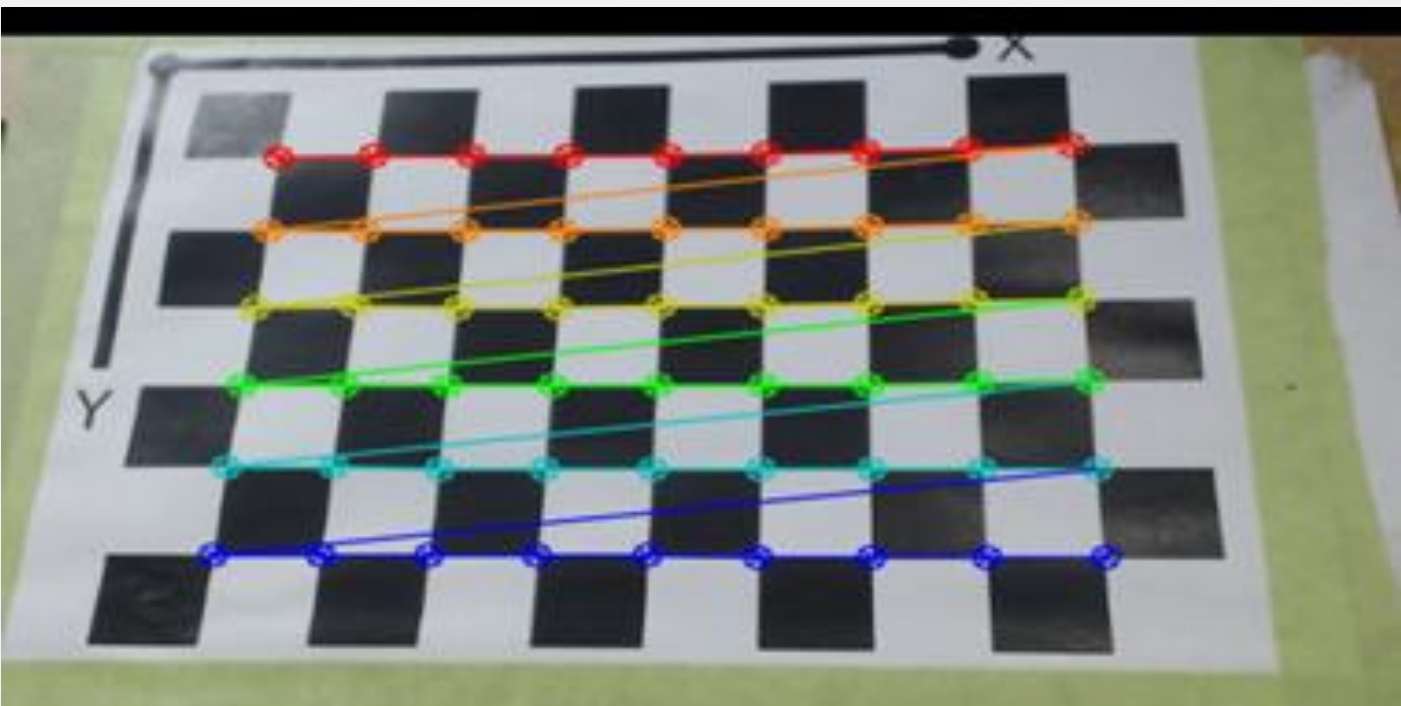


Figure 2: Ultrasound Sequence to 3D Volume Flow

Calibration Module

Stylus Calibration



- Checkerboard corner detection
- Camera parameters estimated
- Lens distortion corrected

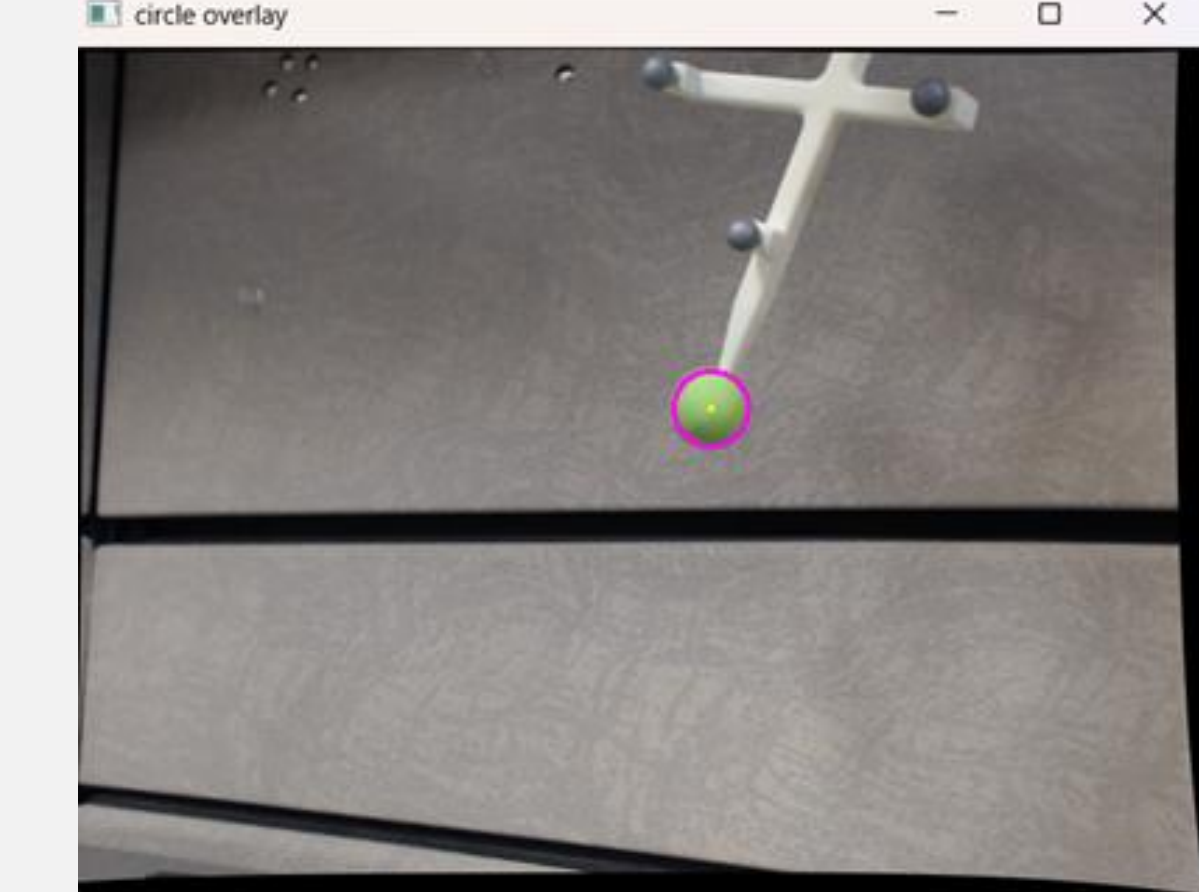
Axial Calibration

- This method identifies the center of the stylus ball by rotating it around a fixed point.

Pivot Calibration

- Estimates offset of tracked stylus body and physical tip by rotation around a fixed point

Hand-eye Calibration



- Computes transform between tracker and camera for accurate live camera-image overlay

ArUco Stylus Tracking

- ArUco markers are detected live to estimate marker pose used for axis rendering and real-time tracking



Results

Results

- Successful .mha generation, validated using 3D Slicer
- Support for multiple I/O devices
- 3D Remodeling of multiple types of scans, such as CT, MRI, ultrasound

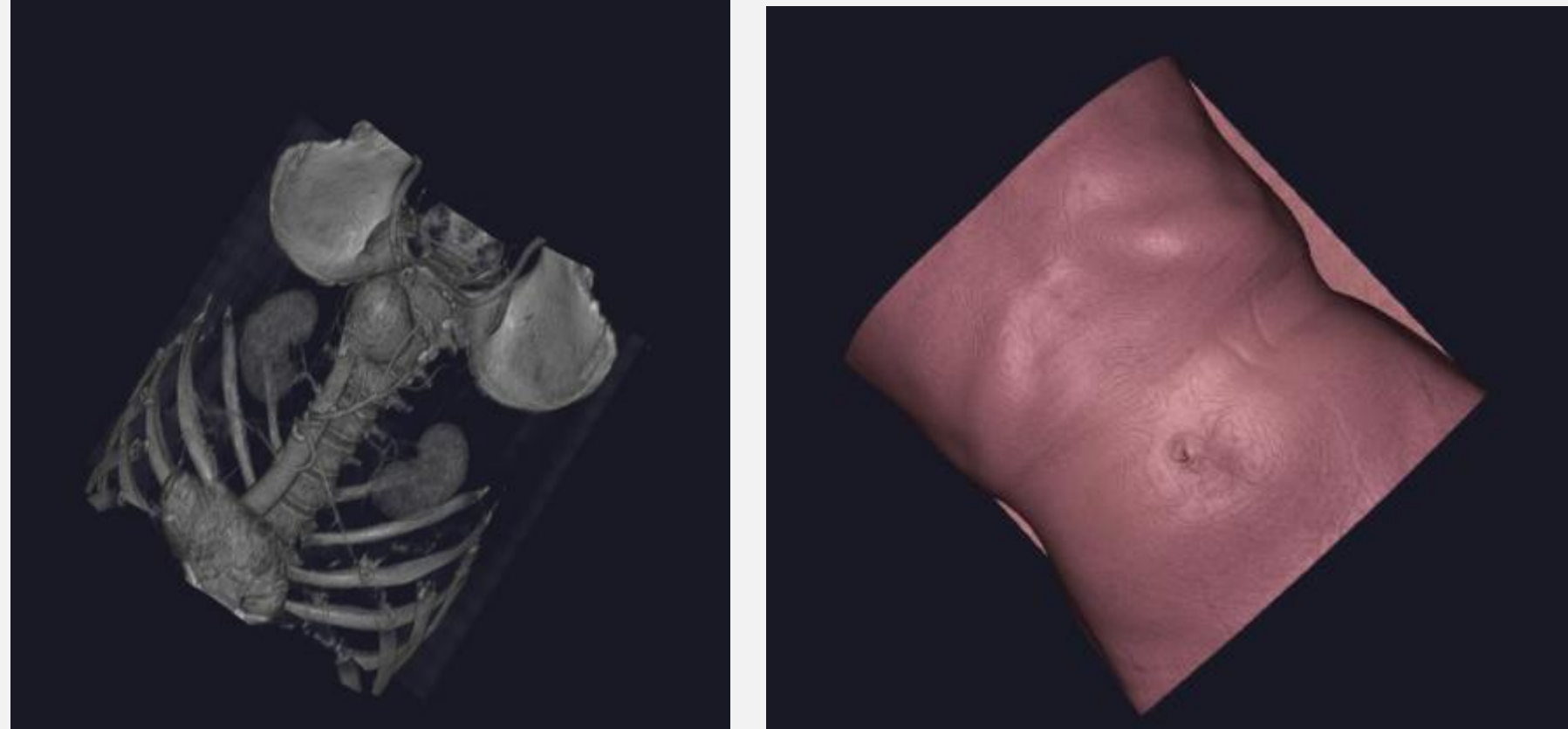


Figure 3: Torso CT Scan Rendering

- Transfer functions to apply colour filters and density scaler

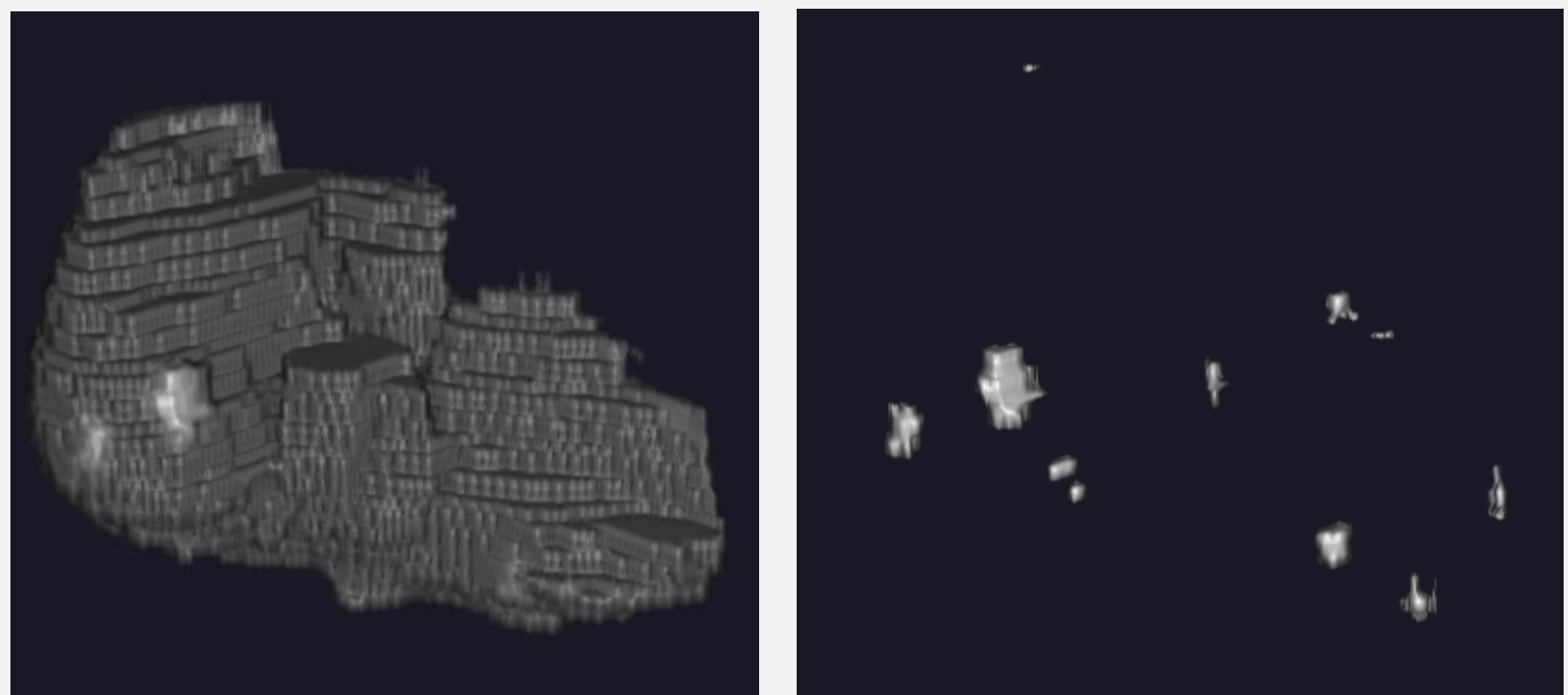


Figure 4: Liver with Tumours Transfer Function

Next Steps & Future Directions

- **Refine Transform Matrix:** Precise calibration for spatial accuracy.
- **Integrated AR View:** Calibrated stylus and volume tracking combined.
- **Dynamic Target Visualization:** Real-time stylus guidance relative to anatomy.
- **Clinical Accuracy Studies:** Performance validation in phantom models.

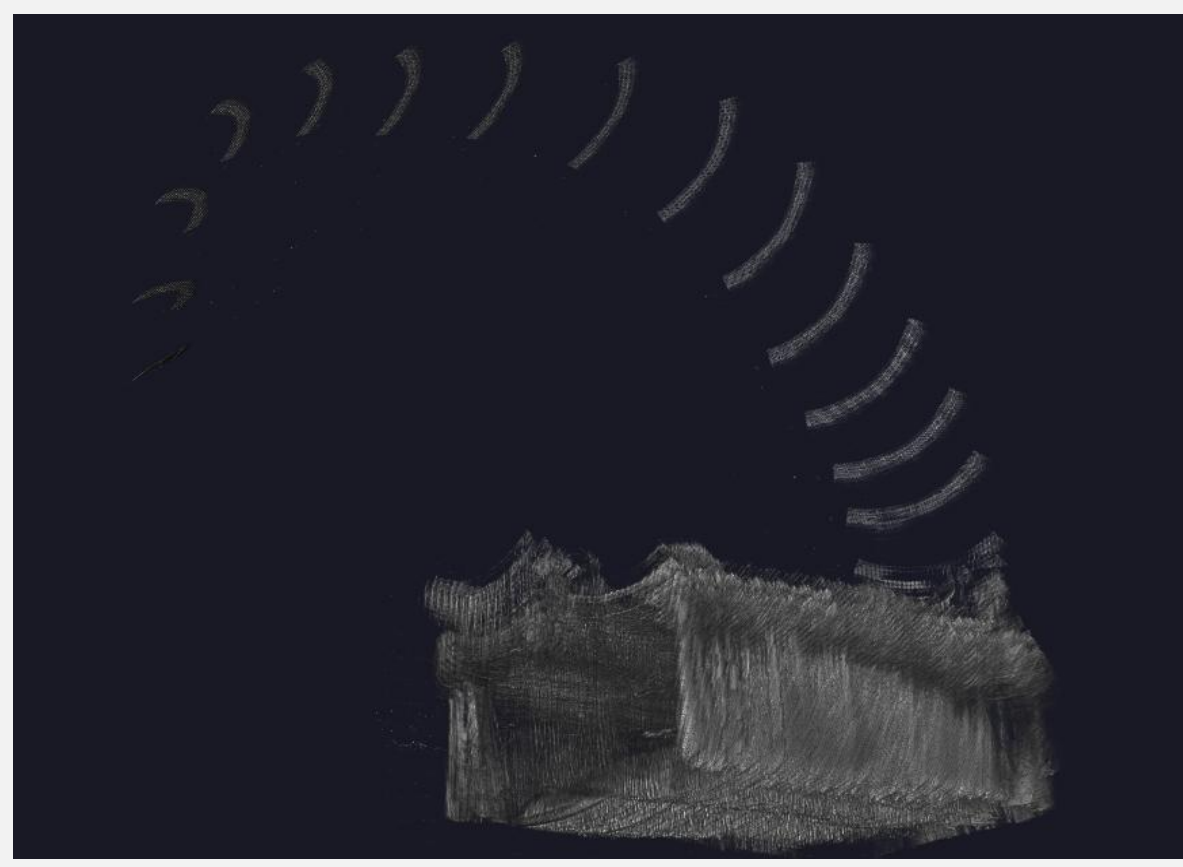


Figure 5: Phantom Scan Recorded Quickly